

PRITHVI SHETTY

Senior Data Engineer | Azure | Databricks | Spark | Snowflake | AWS
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PROFESSIONAL SUMMARY

Senior Data Engineer with 9+ years of experience designing and operating cloud data platforms across healthcare, insurance, and financial services. Build production ETL/ELT pipelines and analytical data systems using Azure Data Factory, Azure Databricks, Python, PySpark, SQL, Delta Lake, Snowflake, AWS, and Airflow. Experienced with large-scale data processing, incremental pipelines, dimensional modeling, Spark and SQL performance tuning, data quality, governance, and secure handling of regulated healthcare data. Partner with analysts, data scientists, and product teams to translate business requirements into reliable production datasets for analytics, reporting, and machine learning.

SELECTED IMPACT

- Process approximately 500 GB to 1 TB of healthcare and operational data per month through Azure-based production data pipelines.
- Support 15-20 production ETL/ELT pipelines integrating data from approximately 8-12 relational, API, and file-based source systems.
- Improved targeted Databricks and Spark workload runtime by approximately 25-30% through execution-plan analysis, partitioning, join optimization, and caching.
- Reduced recurring downstream data-quality issues by approximately 20-25% through automated schema, reconciliation, and business-rule validation.
- Improved Redshift reporting query performance by 35% at BCBS through workload tuning and warehouse table design.
- Reduced legacy ETL processing time by 30% at Northern Trust through workflow modernization and Informatica-to-Talend migration.

TECHNICAL SKILLS

Cloud & Data Platforms: Azure, AWS, Azure Data Factory, Azure Databricks, ADLS Gen2, Delta Lake, Snowflake, Azure Synapse, Azure SQL Database, AWS Glue, S3, Redshift, EMR, Lambda, PostgreSQL

Programming & Data Engineering: Python, PySpark, Advanced SQL, Apache Spark, Spark Structured Streaming, ETL/ELT, incremental processing, REST API ingestion, Apache Kafka, Apache Airflow, Apache NiFi, Informatica, Talend, Hadoop, Hive

Data Modeling & Warehousing: Dimensional Modeling, Star Schema, Fact and Dimension Modeling, Data Marts, Data Warehouse Design, Delta Lake, Snowflake, Redshift, dbt, Source-to-Target Mapping, Data Contracts, Incremental Pipelines

Performance & Reliability: Spark Execution Plans, Partitioning, Shuffle Optimization, Join Optimization, Caching, Predicate Pushdown, Query Optimization, Schema Validation, Source-to-Target Reconciliation, Monitoring, Alerting, Production Troubleshooting

Governance & Security: Microsoft Entra ID, Azure Key Vault, RBAC, Unity Catalog, Data Lineage, HIPAA-Aligned Data Handling, PHI/PII Protection, IAM, KMS, Encryption

DevOps & Platform Engineering: Terraform, Azure DevOps, Jenkins, Git, CI/CD, Docker, Kubernetes, AKS, EKS, Deployment Automation, Infrastructure as Code

Analytics & ML Enablement: MLflow, Feature Engineering, Predictive Analytics, ML-Ready Data Assets, Power BI, Tableau

PROFESSIONAL EXPERIENCE

CVS Health | Woonsocket, RI | Senior Data Engineer | Apr 2024 - Present

- Designed and productionized Azure data pipelines using ADF, Azure Databricks, Python, PySpark, SQL, Delta Lake, and ADLS Gen2, processing approximately 500 GB to 1 TB of healthcare and operational data per month across relational, API, and file-based sources.
- Built reusable Python/PySpark transformation libraries for schema enforcement, deduplication, standardization, and business-rule processing across 15-20 production ETL/ELT pipelines, reducing development effort for new datasets by approximately 25-30%.
- Implemented incremental ingestion and upsert patterns using watermarks, partition pruning, and Delta Lake merge logic across 8-12 source systems, reducing unnecessary full-history processing by approximately 40-50% on applicable workflows.
- Tuned high-volume Spark and SQL workloads by analyzing execution plans, shuffle behavior, and stage-level bottlenecks and optimizing partition counts, join strategies, caching, and predicate pushdown, improving runtime by approximately 25-30% on targeted Databricks pipelines.
- Designed dimensional data models and curated data marts using fact and dimension structures, surrogate keys, and conformed dimensions to support BI, analytics, and data science workloads with consistent downstream business logic.
- Implemented automated data-quality controls using Python, SQL, and PySpark for schema drift, null thresholds, duplicate detection, referential integrity, source-to-target reconciliation, and business-rule validation, reducing recurring downstream data-quality issues by approximately 20-25%.
- Built metadata-driven ADF orchestration frameworks with parameterized datasets, dependency management, retry policies, failure branches, and centralized logging to reduce hard-coded pipeline logic and improve production recoverability.
- Applied Microsoft Entra ID, Key Vault, RBAC, and governed ADLS/Databricks access patterns to protect PHI/PII and support HIPAA-aligned data handling across development and production environments.

Galderma | New York, NY | Senior Data Engineer | Nov 2023 - Mar 2024

- Supported and enhanced Azure and Databricks healthcare data pipelines integrating clinical, claims, operational, and third-party data into governed analytical layers.
- Built PySpark and Delta Lake transformations for large-scale structured and semi-structured healthcare data used in reporting, predictive analytics, and ML-ready workflows.
- Designed Snowflake and dbt analytical models using fact and dimension structures for Tableau, Power BI, and decision-support datasets.
- Implemented Unity Catalog lineage, access controls, Key Vault-backed security, and automated validation and reconciliation checks for HIPAA-aligned data processing.
- Developed Python and Airflow REST API ingestion patterns and supported CI/CD, monitoring, and production troubleshooting across healthcare data workflows.

Blue Cross Blue Shield | Chicago, IL | Data Engineer | Dec 2021 - Nov 2023

- Built and operated 30+ production batch, streaming, and orchestration pipelines across AWS, Spark, Kafka, Airflow, Snowflake, and Redshift for enterprise insurance analytics.
- Integrated 15+ claims, policy, transaction, and operational systems using reusable source mappings, data contracts, transformation logic, and validation standards.
- Processed millions of regulated records with Python, SQL, and Spark on AWS EMR for risk scoring, claims forecasting, underwriting optimization, and enterprise analytics.
- Maintained 98%+ pipeline SLA adherence through Airflow dependency management, monitoring, alerting, failure detection, debugging, and production remediation.
- Improved Redshift reporting-query performance by 35% through distribution-key design, sort-key optimization, columnar modeling, SQL tuning, and workload optimization.
- Managed Docker and EKS-based ETL workloads with Terraform-provisioned AWS infrastructure and IAM, KMS, Security Hub, and CloudWatch controls for regulated insurance data.

Northern Trust Bank | Chicago, IL | ETL Engineer | May 2020 - Dec 2021

- Modernized 25+ Informatica and Talend ETL workflows supporting transaction, trade, risk, finance, and regulatory-reporting datasets in a financial-services environment.
- Built AWS Glue, Lambda, S3, Redshift, Airflow, Kafka, and Spark workflows for financial reporting, analytics delivery, and near-real-time operational processing.
- Consolidated transaction, trade, and risk data into 15+ curated datasets using standardized transformation logic, dimensional models, validation, reconciliation, and reporting rules.
- Reduced legacy ETL processing time by 30% by leading an Informatica-to-Talend migration and refactoring high-volume transformation workflows.
- Improved production reliability through validation and reconciliation controls plus Jenkins CI/CD, Terraform, Docker, CloudWatch monitoring, logging, and alerting.

Luxoft India Pvt. Ltd. | India | Hadoop Engineer | Jul 2017 - Jul 2018

- Built Hive, Spark, Python, and SQL ETL pipelines and administered Hadoop clusters supporting distributed storage, scalable batch processing, and enterprise datasets.
- Improved throughput across 20+ Hadoop and Spark jobs through partitioning, compression, query optimization, and resource tuning and automated deployments through Jenkins and Docker.

Optimum Info System Pvt. Ltd. | India | Data Analyst | Nov 2015 - Jun 2017

- Automated data extraction, cleansing, transformation, validation, and reporting using Python, pandas, NumPy, and SQL for analytics-ready datasets.
- Delivered 10+ Tableau dashboards and optimized SQL queries by translating stakeholder requirements into measurable KPIs and structured reporting workflows.

EDUCATION

California State University Northridge | Master of Science in Engineering Management | 2018 - 2020**CERTIFICATIONS**

Preparing for Microsoft Certified: Azure Data Engineer Associate | Preparing for Databricks Certified Data Engineer Associate